CANDIDATE NAME ___________________________ CG _________

SERANGOON JUNIOR COLLEGE
JC2 PRELIM PRACTICAL EXAMINATION 2017

H2 BIOLOGY ANSWER SCHEME 9744

2.5 hours

Question 1

You are provided with a quantity of vitamin C solution and a dye called DCPIP.

You are also provided with three test-tubes containing respectively lemon juice, orange juice, grapefruit juice, and labelled as such. These juices contain natural vitamin C and the dye DCPIP can be used to determine the concentration of this vitamin in the juices.

Apparatus: 6 test-tubes and a test-tube rack
4 plastic teat pipettes
Plastic ruler

Material: 30 cm^3 DCPIP solution, labelled '**DCPIP**'
50 cm^3 vitamin C solution, labelled '**Vitamin C solution**'
50 cm^3 lemon juice
50 cm^3 orange juice
50 cm^3 grapefruit juice

Proceed as follows:

1. Into a clean test-tube, transfer a quantity of the dye DCPIP to a depth of 0.5 cm. Take note its colour.

2. Fill a teat pipette with vitamin C solution. Add one drop of vitamin C solution to the DCPIP solution in the test-tube and shake gently. Continue to add the drops, counting the number of drops which are needed to bring about a colour change. Shake gently after each drop, refilling the pipette if necessary.

3. Record the initial colour of DCPIP (from step 1) and the first colour change after vitamin C is added as well as the number of drops counted to bring about this colour change in a suitable table.

4. After the first colour change, continue adding drops of vitamin C and counting the drops until the DCPIP solution becomes colourless/or consistent pale yellow. (Ignore any coloured granules that might form.). Record the number of drops counted in the same table from step 3.

5. Repeat steps 1 to 4 adequately to obtain enough data for analysis, cleaning all apparatus before use.

6. Place the DCPIP solution into each of three clean test-tubes to a depth of 0.5 cm. (The amount of DCPIP solution must be exactly the same in each of the tubes). Label the tubes A, B and C.

7. Fill a clean teat pipette with lemon juice and drop by drop add this to the contents of tube **A**, shaking the tube gently after each drop. Count the number of drops needed to turn the DCPIP solution colourless. Repeat this step adequately to obtain enough data for analysis.

8. Repeat the step 7 with orange juice and grapefruit juice, using a clean pipette each time to add the juice to the DCPIP solution in tubes **B** and **C** respectively.

9. Record the results for the three juices in an appropriate table. [5]

Solution	**Replicates**	**Initial colour of DCPIP**	**First colour change**	**Drops required to achieve 1st colour change**	**Second Colour Change**	**Drops required to achieve 2nd colour change**
Vitamin C	1	Blue	Brown/Yellow	1	Pale Yellow	2-5
	2					
	Average					
Lemon Juice						
Grapefruit juice						
Orange juice						

Table with headings – 1
Results for Vitamin C with at least one replicate – 1
Results for Juices with at least one replicate – 1
Trend for Juices with at least one replicate – 1 (Lowest drops (Orange) to highest drops (Lemon)
Calculation of average for each solution - 1

10. What conclusions can you draw from your results? [3]

- Relate trend to relative concentrations of Vitamin C
- Higher concentrations of Vitamin C = lower pH = more H+
- DCPIP reduced faster = faster colour change

11. Comment on the main source(s) of error and the limitations of the measurements or experimental procedure. [3]

identifies three of:

- *uneven mixing of the juice with DCPIP solution;*
- *measuring DCPIP to a depth of 0.5 cm in test-tube;*
- *inconsistent drop size and release of drop from teat pipette;*
- *use of ruler to mark out a depth of 0.5 cm on test-tube;*
- *visual comparison of colour change (when DCPIP decolourises);*
- *varying drop size of vitamin C (or juice) due to way and the speed at which it is released from teat pipette;*
- *viscosity of fruit juice affecting size of the drop being formed before it is released from the teat pipette;*

12. What improvements could you make to the experimental procedures to overcome these sources of error? [3]

Shake before using juice
Use colourimeter to determine colour change
use a syringe / graduated pipette to transfer a known volume of DCPIP;
use a burette to dispense equal sized drops / regulate the release the fruit juice;

[Total: 15 marks]

Question 2

You are provided with a leaf labelled Q. In this question you will be required to investigate the number of stomata present on the lower surface of leaf Q.

Proceed as follows:

1. Use nail varnish to cover a small area of the lower epidermis of leaf Q. Apply a thin layer over an area about $1cm^2$. Avoid any large veins that may be present.
2. Repeat this process for three different areas of the lower epidermis.
3. Allow the nail varnish about 20 minutes to dry. (Proceed to question 3 while the leaf is drying)
4. After 20 minutes, use a razor blade or a fine scalpel to lift one edge of the layer of nail varnish. Use forceps to then gently peel a layer of nail varnish off the leaf.
5. Transfer this layer of nail varnish to a slide and cover with a cover slip.
6. Examine the slide using a microscope and count the number of stomata you can observe in the field of view.
7. State which objective you chose to make this stomatal count and explain your choice. [1]

X10. Idea that number of stomata is countable, not too many, not too few, just nice.

8. Repeat the counting of stomata in the field of view for a second piece of peeled nail varnish. Calculate the mean number of stomata per field of view in the space below. [2]

Working - 1
Precision 3sf - 1

9. Make a high power drawing of three adjacent stomata from either of your stomatal peels. [3]

Accurate Drawing of three adjacent stomata – 1
Accurate drawing of epidermal cells between stomata (Shape hexagonal plus cell wall as double line) - 1
Label of guard cell and stoma – 1

10. Using the eyepiece graticule in your microscope and the provided stage micrometer, find the actual length, in µm, of one of the guard cells that you have drawn. [3]

State objective used -1
Number of eyepiece graticule x calibrated value for that objective – 1
Final answer to 3sf in micrometer - 1

(40x – 30/35 eyepiece graticule – calibrated value = 0.0025, answer = 75.0µm)

Objective Lens	**X4**	**X10**	**X40**	**X60**
Diameter of Field of View (mm)	5	2	0.5	0.333
Length of one eyepiece division (mm)	0.025	0.01	0.0025	0.00167

A researcher obtained leaves from two different plants, Plant A and Plant B. From the same forest. He found that the number of stomata on the underside of the both leaves differed. He ensured that he calculated the number of stomata based on per unit area and ensured he had sufficient replicates. However, the numbers still did not match. The mean density of stomata per $1cm^2$ for leaves A and B were as follows:

Leaf A – 234
Leaf B – 297

11. While the researcher expected a difference in the number of stomata, he could not be sure if the difference seen was significant. Suggest a statistical test he could use to confirm if the difference was significant. [1]

T-Test

12. When carried out this test, the probability value he obtained was less than 0.05. Comment on what these results show and suggest an explanation for the pattern seen in Leaves A and B collected by the researcher. [4]

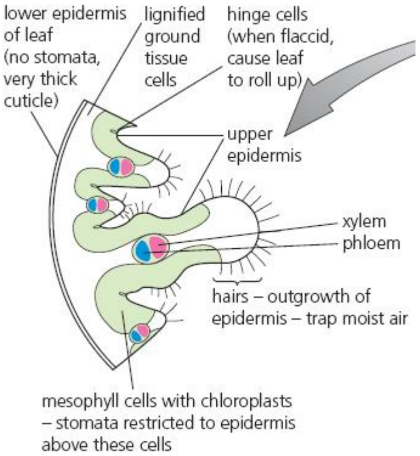
- **Shows that the probability that the difference in density of stomata between the two leaves being due to chance is less than 5%.**
- **The difference is therefore statistically significant.**
- **Leaf A may be found in sunny areas while leaf B is in the shade/ Leaf A plant may be growing in low water content soil while leaf B plant is growing in more waterlogged soil/difference in carbon dioxide levels/AVP**
- **Ref to idea that higher stomatal density lead to increased water loss or increased carbon dioxide uptake and vice versa.**

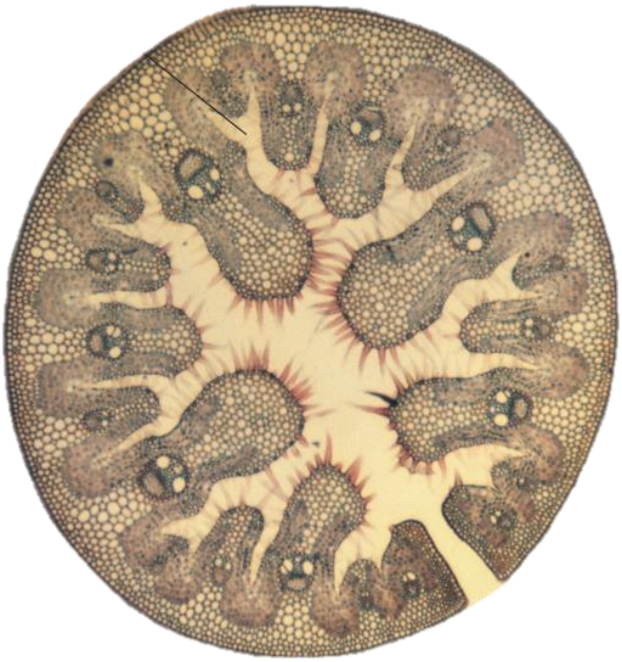
[Total: 14 marks]

Question 3

Slide S1 is a tranverse section from a leaf of the *Amophilia* plant.

(a) In the space below, draw a labelled plan drawing of this leaf. [6]



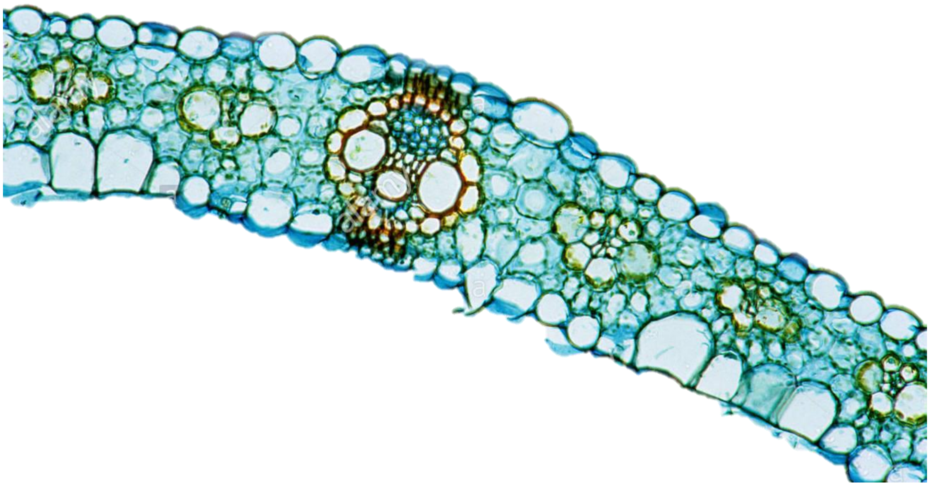


Title – 1 <u>(Plan drawing of *Amophillia* Leaf (T.S.) X40/X100)</u>
Accurate size and shape of regions - 1
Mesophyll Labelled – 1
Vascular Bundle (with xylem and phloem) Labelled – 1
Outer epidermal layer represented with a double line – 1
Calculation of magnification of drawing - 1

(b) Using the provided **slide graticule**, measure the diameter of the leaf. You may assume that the slide graticule can be used as you would use a ruler. [1]

Diameter of leaf: 1.90 to 1.95 mm

The following **Figure 2.1** is a microscope image of a section of a typical monocot leaf.



From alamy.com

Figure 2.1

(c) Given that the magnification of Figure 2.1 is 50X, calculate the actual width of the specimen. [2]

Working - 1
Precision 3sf - 1

(d) Identify three differences between a typical monocot leaf and the *Amophilia* leaf. [3]

In *Amophillia*

- **Presence of epidermal hairs**
- **Folding/Rolling of leaf**
- **Sunken/hidden stomata (idea)**
- **No obvious guard cells**
- **AVP**

Any 3

(e) Suggest reasons for the differences you have identified. [2]

- **Plant found in hot and dry regions**
- **Lacking in water**
- **Avoid excessive loss of water from evapo-transpiration**

Any 2

[Total: 14 marks]

Question 4

The Tasmanian Tiger or Thylacine (*Thylacinus cynocephalus*), the world's largest carnivorous marsupial, was once common throughout Australia and Papua New Guinea. The Thylacine resembled a large, short-haired dog with a stiff tail which smoothly extended from the body in a way similar to that of a kangaroo. The female Thylacine had a pouch with four teats, but unlike many other marsupials, the pouch opened to the rear of its body.

An example of convergent evolution, the Thylacine showed many similarities to the members of the Canidae (dog) family of the Northern Hemisphere: sharp teeth, powerful jaws, raised heels and the same general body form.

Due to human activities, the Thylacine was hunted to extinction by early 1930. A Thylacine specimen with soft tissue remaining is found in the Australian Museum in Sydney.

Imagine that you are a researcher in the Australian Rare Fauna Research Association. Recently it was reported that locals near Mount Carstensz in Western New Guinea had sighted creatures that resemble Thylacines. Some members of the Association believe that the creatures sighted may be descendents of the Thylacine, while other members believe that the creatures may be a new species. If the former were true, then the Thylacine is not extinct and conservation effects may revive the species.

Plan an investigation to investigate if these creatures found in Western New Guinea are descendents of the Thylacine that was thought to be extinct.

Your planning must be based on the assumption that you have been provided with the following equipment and materials.

- Tissue sample from the museum specimen and from the Western New Guinea creatures under investigation
- Pestle and mortar
- DNA extraction buffer solution
- Microcentrifuge tubes
- Centrifuge
- Restriction enzymes
- Agarose or polyacrylamide gel plate
- Suitable source of electric current
- Radioactive probe
- Nitrocellullose membrane
- Autoradiography equipment

Your plan should have a clear and helpful structure to include:

- An explanation of the theory to support your practical procedure
- A description of the method used including scientific reasoning behind the method
- The type of data generated by the experiment
- How the results will be analysed including how the origin of the organism can be determined

[Total: 14]

1. **Theoretical consideration or rationale of the plan to justify the practical procedure;**
2. **Method of <u>DNA extraction</u> including <u>homogenization</u> and use of <u>buffers;</u>**
3. **Preparation of samples for electrophoresis, including <u>use of centrifuge</u>**
4. **Selection of restriction enzyme and <u>reasons for the selection;</u>**
5. **Amplification of <u>DNA fragments</u> using PCR including <u>detail of PCR</u>;**
6. **Separation of fragments by <u>gel electrophoresis</u> and the <u>principles</u> behind the separation;**
7. **<u>Transfer</u> of DNA onto nitrocellulose membrane;**
8. **<u>Hybridization</u> with <u>radioactive labeled DNA probe;</u>**
9. **<u>Autoradiography</u> method;**
10. **Method of band visualization, e.g. exposure to X-ray or chemiluminescent substrate;**
11. **Significance of <u>matching bands</u>;**
12. **The correct use of technical and scientific terms; valid safety concerns**

1. **Theoretical consideration or rationale of the plan to justify the practical procedure;**

    - **Homologous regions of DNA** in the Thylacine and the creatures found in Western New Guinea have **variable number of tandem repeats (VNTR).**
    - Restriction digestion of homologous regions produces **restriction fragments of different lengths,** the animals exhibit **Restriction Fragment Length Polymorphism (RFLP).**
    - **Comparison of the DNA banding patterns from** Thylacine with the creatures found in Western New Guinea **will determine if they are the same species.**

2. **Method of <u>DNA extraction</u> including <u>homogenization</u> and use of <u>buffers;</u>**

    - Add **<u>DNA extraction buffer</u> to tissue sample before homogenization**
    - **Mechanically break tissue samples** during **homogenization** using **<u>mortar and pestle,</u>** to release DNA

3. **Centrifuge homogenized samples in a <u>microcentrifuge tube</u>** to **separate DNA** from the **rest of the cellular debris**.

    - Use a **<u>micropipette</u>** to transfer the **supernatant** which contains the DNA into another clean microcentrifuge tube.
    - Add **<u>ice-cold ethanol</u>** to **precipitate the DNA out of solution**.

4. **The <u>same</u> restriction enzyme,** for example, **EcoRI, which is used to cleave DNA from the** Thylacine and the creatures found in Western New Guinea **<u>at different restriction sites</u>**, producing **<u>differentiating number and length of restriction fragments</u> between the species**.

5. The **purified DNAs from both specimens are subjected to Polymerase Chain Reaction** (PCR) in a **<u>thermocycler</u>** to **amplify** a particular region of the DNA using ***<u>Taq</u>* <u>polymerase</u>**.

   - **One PCR cycle** consists of **3 steps:**
     - **Denaturation of DNA molecules at (<u>95ºC</u>), to break the hydrogen bonds** so that the **double-stranded DNA separates into single strands.**
     - **Annealing of Primer to DNA strands at (<u>55ºC</u>) to complementary sequences flanking the target sequence to be amplified**
     - **Extension step at (<u>72ºC</u>) for Taq polymerase to catalyse the synthesis of a complementary strand of DNA for each of the single strands of the target DNA.**

6. **Separation of fragments by <u>gel electrophoresis</u> and the <u>principles</u> behind the separation;**

   - **Mix DNA fragments after restriction digestion with <u>loading dye</u>.**
   - **Load the DNA samples** from the 2 animals **into <u>separate wells</u> of the <u>agarose gel</u>** to perform **gel electrophoresis at <u>100 V for 30 min</u>.**
   - **Load DNA ladder onto a separate well** as a **reference for identification of band sizes.**
   - Gel electrophoresis separate DNA fragments **based on size - Smaller size fragments migrate more rapidly/at a faster rate than large fragments to the positive end.**
   - DNA is **negatively-charged** and will **migrate to the positive electrode.**

7. **<u>Transfer</u> of DNA onto nitrocellulose membrane;**

   - Restriction fragments in the agarose gel are transferred to a piece of **<u>nitrocellulose membrane</u> by capillary action during <u>Southern Blotting</u>.**
   - The gel is immersed in **alkaline solution (sodium hydroxide)** to **denature double-stranded DNA into single strands.**

8. **<u>Hybridization</u> with <u>radioactive labeled DNA probe;</u>**

   - The nitrocellulose membrane is then incubated with a **<u>radioactively labelled, single-stranded DNA probe</u>** which is **<u>complementary and thus hybridised to the target VNTR sequences</u>**, via formation of **hydrogen bonds**. Excess probe was washed off.

9. **<u>Autoradiography</u> method;**

   - **Detect the position** of the hybridised probes by **<u>autoradiography</u>.** A sheet of **<u>X-ray film</u> is <u>placed over the nitrocellulose membrane</u>.**

10. **Method of band visualization, e.g. exposure to X-ray or chemiluminescent substrate;**

- **Bands of radioactivity will be detected on the X-ray film** where the **radioactively labelled probe hybridised to complementary VNTR sequences.**

11. **Significance of <u>matching bands</u>;**

- **Compare DNA banding patterns** from Thylacine with the creatures found in Western New Guinea.
- **Identify bands that** creatures found in Western New Guinea **have in common/match** with the Thylacine.
- Presence of a **high number of common or matching bands at the chosen RFLP loci** indicates a **high possibility** that creatures found in Western New Guinea are **evolutionarily related** to and are likely to be the **descendents of the Thylacine**. **If the banding patterns are different, then they are not the same species.**

12. **Valid safety concerns**

- **<u>Sodium hydroxide</u> can be an <u>irritant</u>, avoid direct contact with hands and wear gloves.**
- **<u>Risk of electrocution</u>, do not switch on/off power socket and power pack of gel electrophoresis** with **wet hands to avoid being electrocuted**. Ensure **hands are dry** when using appliances to prevent electrocution.
- **<u>Probes are radioactive</u>, wear protective gear and gloves when handling probes during southern blotting and autoradiography and work behind a radioactive shield.**
- **<u>Ethidium bromide</u> is a <u>carcinogen</u>. Handle gel using protective gloves.**

End of Paper